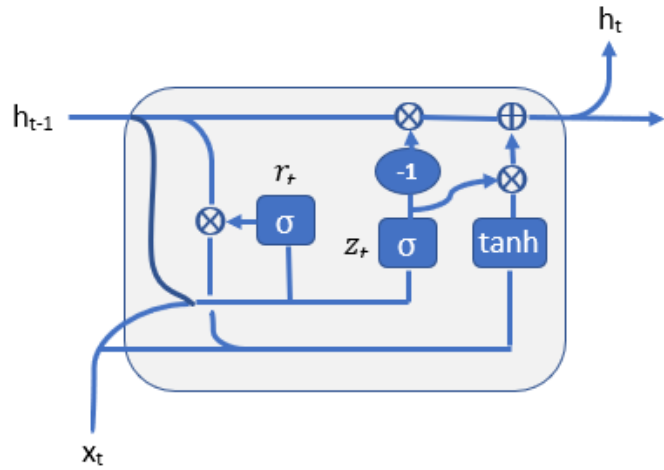


Gated Recurrent Unit (GRU)

Unlike LSTM, GRU has only one State and only two Gates

Hidden State
&
Update Gate and Reset Gate

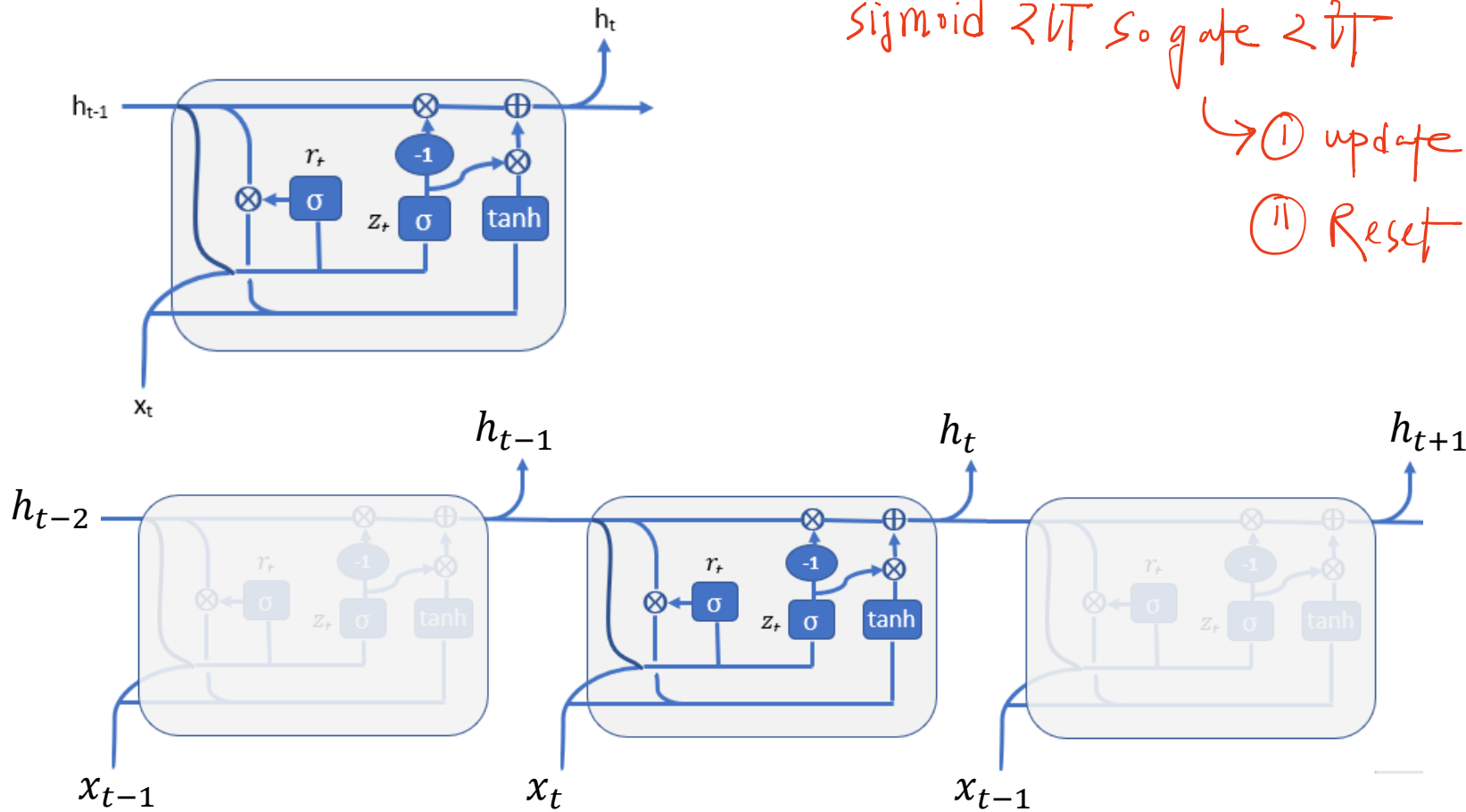
Gated Recurrent Unit (GRU)



Gated Recurrent Unit (GRU)

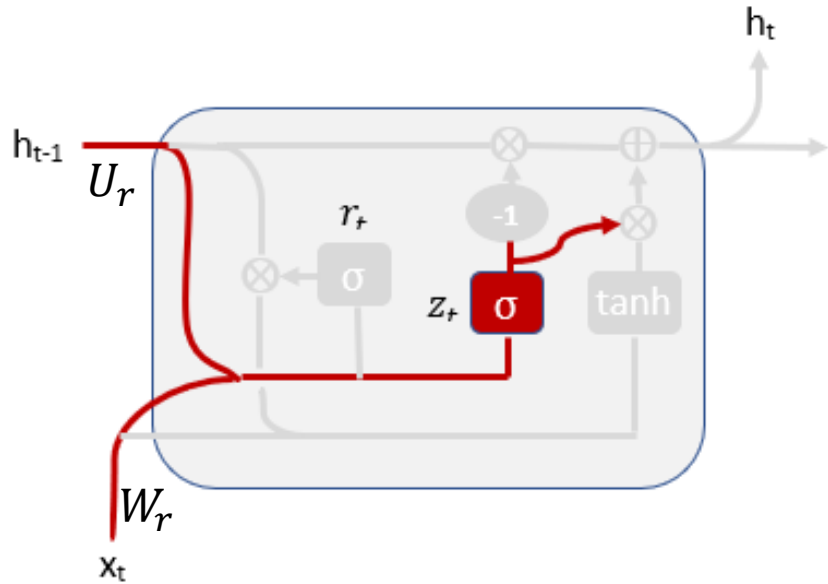
sigmoid $\hat{z}_t$ so gate $\hat{z}_t$

*① update gate
② Reset gate*



Update Gate

How much of the past information need to be considered to the next time step.

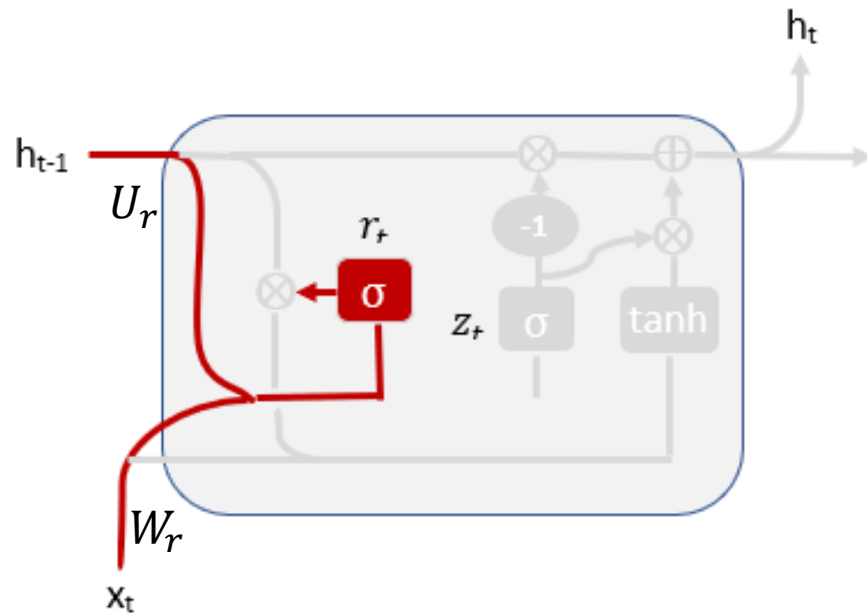


$$Z_t = \sigma(\underbrace{W_Z \cdot x_t}_{\text{input token}} + \underbrace{U_Z \cdot h_{t-1}}_{\text{prev context}})$$

linear combination

Reset Gate (r_t)

How much of the past information to be forgotten?



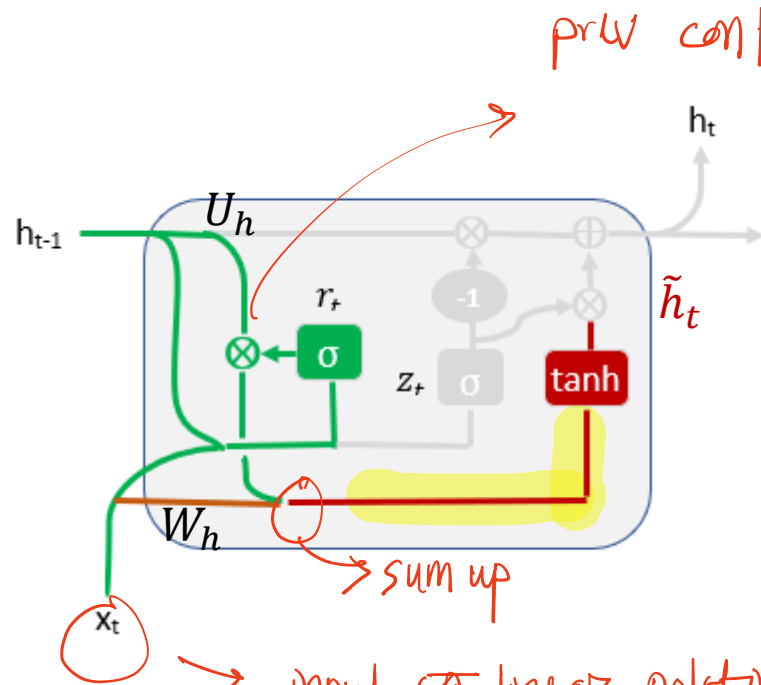
percentage $\downarrow$ prev context $\downarrow$

$$r_t = \sigma(W_r \cdot x_t + U_r \cdot h_{t-1})$$

$\uparrow$ not taken

linear polation

Internal Information (Current Memory Content)



prev context এর কিছু অংশ ভুলে যাচ্ছি (Reset gate)

$$\tilde{h}_t = \tanh(W_h \cdot x_t + r_t \otimes U_h \cdot h_{t-1})$$

Consider the present input and useful information of the previous hidden state to produce new internal information

→ input এর linear polatim দিয়ে tanh এ পাঠাই

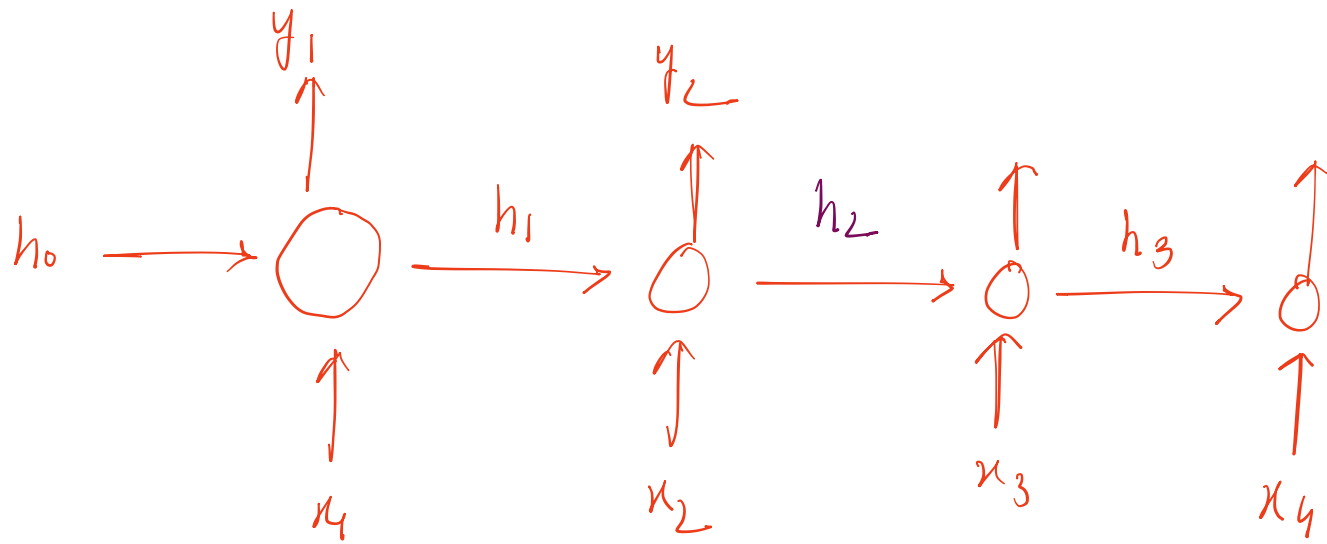
→ নতুন info create

update gate ০.১

→ নতুন info ০.১ রাখবে

৯০% পুরোনো info রাখবে

নতুন hidden context



I bought a mouse and it is black. → computer ও mouse নাকি animal mouse?
next line দেখা লাগবে,

next line → It supports USB

→ named entity target এর জন্য future token এর উপর ও dependency থাকে।

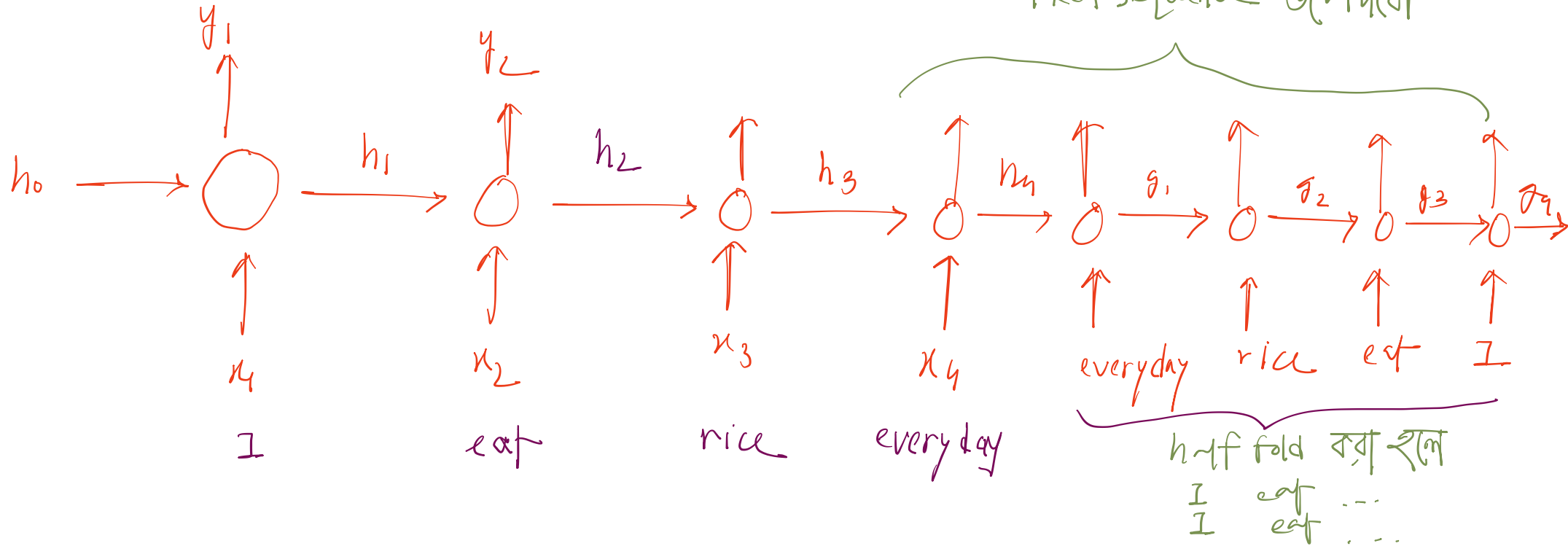
RNN এ future info দেওয়ার সুযোগ নেই।

→ কোন parts of speech, noun টা কিনির্দেশ করতে। শূন্য Current token দেখে বোঝা যায় না।

I eat rice everyday.

Bi-RNN → Bidirectional RNN

token sequence দেয়া দিবে



৭৭ এ যখন জায়গি → ঘামনে পিছনে দুই-পাশের token দেখে ফেলছি

RNN কে sequence কে extend করছি। Output predict করতে কিভাবে? ২ টি hidden context থাকবে তাদের concat.

guaranteed full seq length পাঠ্য তখন use করা যাবে।

classification এ use করা যাবে।

→ sentiment analysis

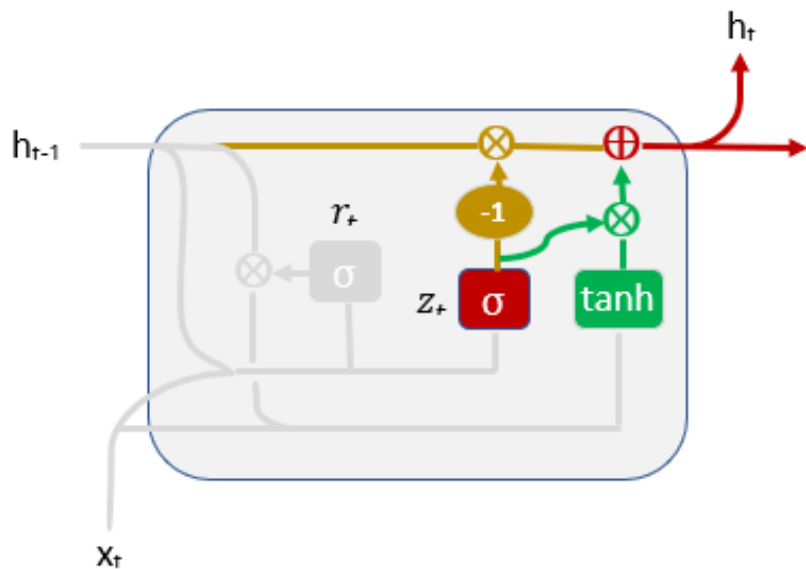
Live transmission হলে use করা যাবে না।

Sarcasm detection — BiRNN use

→ memory বেশি লাগবে।

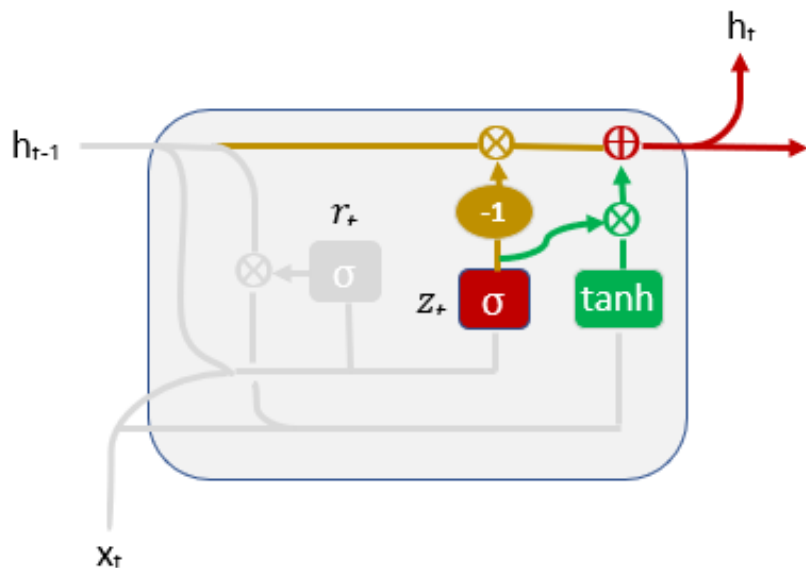
② Inference time এও full seq available থাকবে
যেটা guarantee থাকা লাগবে।

New Hidden State (h_t)



$$h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes \tilde{h}_t$$

New Hidden State (h_t)

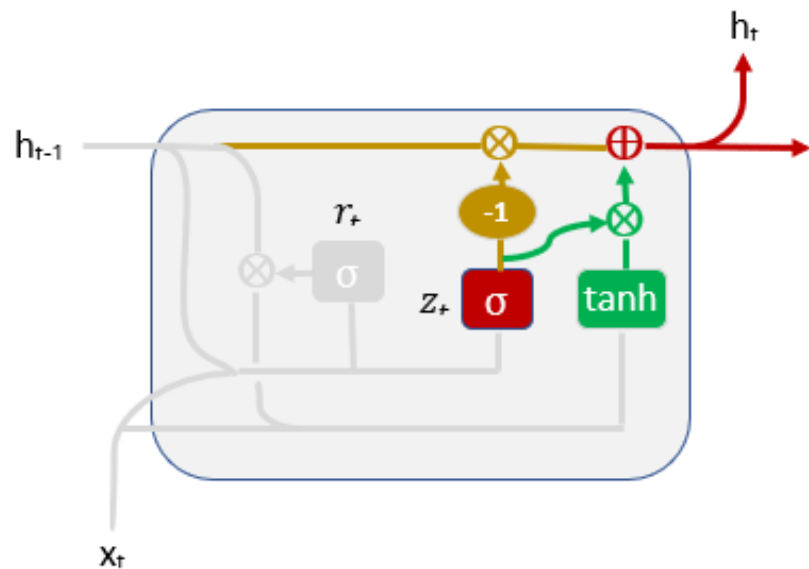


$$h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes \tilde{h}_t$$

Previous hidden state at the position which are ignored in internal information

Useful information from the internal information

New Hidden State (h_t)



$$h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes \tilde{h}_t$$

$$\begin{bmatrix} 1 \\ 0 \\ 0.3 \\ 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 5 \\ 8 \\ h \\ i \\ j \end{bmatrix} \oplus \begin{bmatrix} 0 \\ 1 \\ 0.7 \\ 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}$$

Summary

One State

- Hidden State $h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes \tilde{h}_t$

Two Gates

- Update Gate $Z_t = \sigma(W_z \cdot x_t + U_z \cdot h_{t-1})$
- Reset Gate $r_t = \sigma(W_r \cdot x_t + U_r \cdot h_{t-1})$